Raspberry Pi Pico Troubleshooting Guide

Serial Monitor Issues

1. USB Connection Check

- Ensure the Pico is connected via USB cable (not just power)
- Try different USB cables and ports
- Check if the Pico appears as a device in Device Manager (Windows) or `lsusb` (Linux)

2. Serial Port Configuration

```c
// Make sure you have this in your CMakeLists.txt:
pico_enable_stdio_usb(your_target_name 1)
pico_enable_stdio_uart(your_target_name 0)

// In your main code:
#include "pico/stdio.h"

int main() {
    stdio_init_all();
    // Wait for USB connection (optional)
    sleep_ms(2000);

    printf("Hello from Pico!\n");
    // Your main code here
}
```

3. Reset Sequence

1. Hold BOOTSEL button on Pico
2. Press and release RESET (or unplug/replug USB)
3. Release BOOTSEL
4. The Pico should appear as a mass storage device
5. Flash your .uf2 file
6. Open serial monitor after flashing

Display Issues

1. Check Wiring

Verify connections for common displays:

SSD1306 OLED (I2C):

- VCC → 3.3V (pin 36)
- GND → GND (pin 38)
- SCL → GP1 (pin 2) or your configured SCL pin
- SDA → GP0 (pin 1) or your configured SDA pin

LCD 16x2 (I2C with PCF8574):

- VCC → 5V (pin 40) or 3.3V (pin 36)
- GND → GND
- SCL → GP1
- SDA → GP0

2. I2C Scanner Code

Use this to detect I2C devices:

```c
```

```c
#include "pico/stdlib.h"
#include "hardware/i2c.h"
#include "pico/stdio.h"

int main() {
    stdio_init_all();
    sleep_ms(2000);

    // Initialize I2C
    i2c_init(i2c0, 100000); // 100kHz
    gpio_set_function(0, GPIO_FUNC_I2C); // SDA
    gpio_set_function(1, GPIO_FUNC_I2C); // SCL
    gpio_pull_up(0);
    gpio_pull_up(1);

    printf("I2C Scanner Starting...\n");

    while (1) {
        printf("Scanning I2C bus...\n");

        for (int addr = 0x08; addr < 0x78; ++addr) {
            uint8_t rxdata;
            int ret = i2c_read_blocking(i2c0, addr, &rxdata, 1, false);

            if (ret >= 0) {
                printf("Found device at address 0x%02X\n", addr);
            }
        }

        sleep_ms(5000);
    }
}
```

3. Common Display Addresses

- SSD1306 OLED: 0x3C or 0x3D
- PCF8574 (LCD backpack): 0x27, 0x3F, or 0x20

Debugging Steps

1. Basic LED Test

Start with a simple LED blink to verify the board is working:

```c
```

```
#include "pico/stdlib.h"

int main() {
    const uint LED_PIN = PICO_DEFAULT_LED_PIN;
    gpio_init(LED_PIN);
    gpio_set_dir(LED_PIN, GPIO_OUT);

    while (true) {
        gpio_put(LED_PIN, 1);
        sleep_ms(250);
        gpio_put(LED_PIN, 0);
        sleep_ms(250);
    }
}
```

2. Serial Output Test

```c
#include "pico/stdlib.h"
#include "pico/stdio.h"

int main() {
    stdio_init_all();
    sleep_ms(2000); // Wait for USB connection

    int counter = 0;
    while (true) {
        printf("Hello World! Counter: %d\n", counter++);
        sleep_ms(1000);
    }
}
```

3. Check CMakeLists.txt

Ensure your CMakeLists.txt includes:

```cmake
```

```cmake
cmake_minimum_required(VERSION 3.13)
include(pico_sdk_import.cmake)
project(your_project)

pico_sdk_init()

add_executable(your_target
    your_main.c
    # other source files
)

target_link_libraries(your_target
    pico_stdlib
    hardware_i2c
    hardware_gpio
    # other libraries you need
)

pico_add_extra_outputs(your_target)
pico_enable_stdio_usb(your_target 1)
pico_enable_stdio_uart(your_target 0)
```

Common Solutions

Serial Monitor Not Working:

1. Try different baud rates (115200 is common)
2. Use different serial monitor software (PuTTY, Arduino IDE, VS Code with serial monitor extension)
3. Check if Windows needs USB Serial driver
4. Try resetting the Pico while serial monitor is open

Display Not Working:

1. Verify power supply (some displays need 5V, others 3.3V)
2. Check pull-up resistors on I2C lines (usually built into Pico)
3. Try different I2C pins
4. Verify display library initialization
5. Use I2C scanner to confirm device is detected

Build Warnings:

The warnings you see are generally harmless but can be reduced by:

1. Using the latest Pico SDK version

2. Updating your compiler toolchain
3. Adding appropriate compiler flags in CMakeLists.txt

If none of these steps resolve your issues, please share:

1. Your main source code
2. Your CMakeLists.txt file
3. The specific display/components you're using
4. Your wiring setup